

Drew Schreiner, Ph.D.

Duke University | drew (dot) schreiner (at) duke (dot) edu

EDUCATION

University of California San Diego

2015 – 2021

Ph.D. Experimental Psychology; Cognitive and Behavioral Neuroscience Specialization

University of Colorado at Boulder

2009 – 2013

B.A. Molecular, Cellular, and Developmental Biology

B.A. Psychology with focus in Neuroscience, *magna cum laude*

RESEARCH EXPERIENCE

Postdoctoral Scholar

2021 – Present

Duke University. Advisor: Dr. Richard Mooney

Graduate Student Researcher

2015 – 2021

University of California San Diego. Advisor: Dr. Christina Gremel

Laboratory Technician

2015

Science Care, Aurora, Colorado

Professional Research Assistant

2013 – 2014

University of Colorado at Boulder. Advisor: Dr. Ryan Bachtell

Undergraduate Research Assistant

2011 – 2013

University of Colorado at Boulder. Advisor: Dr. Ryan Bachtell

GRANTS AND FELLOWSHIPS

NIH K99/R00 Pathway to Independence Award (K99 NS144525) – \$966,000

2025 – 2030

NIH F32 NRSA BRAIN Initiative Postdoctoral Fellowship (F32 MH132152) – \$236,000

2023 – 2025

NIH F31 NRSA Predoctoral Fellowship (F31 AA027439) – \$75,000

2019 – 2021

Society for Neuroscience Chapter Grant Award for NeuWriteSD (3 awards) – \$4,500

2018 – 2020

Howard Hughes Medical Institute Biomedical Sciences Individual Grant – \$2,400

2012 – 2013

Undergraduate Research Opportunity Program Individual Grant – \$2,400

2012

Undergraduate Research Opportunity Program Assistantship Grant – \$600

2011 – 2012

AWARDS

Society for Neuroscience Trainee Professional Development Award

2025

Best Postdoc Talk, Duke Neurobiology Retreat

2023

NIH Outstanding Scholar in Neuroscience Award

2023

Society for Neuroscience Trainee Professional Development Award

2022

Research Society on Alcoholism Enoch Gordis Award Finalist

2021

Research Society on Alcoholism Student Merit Award

2021

Cosyne New Attendees Travel Grant

2020

UC San Diego Graduate Student Administration Travel Grant

2020

Research Society on Alcoholism Student Merit Award

2019

Society for Neuroscience Trainee Professional Development Award

2018

Research Society on Alcoholism Student Merit Award

2017

Magna cum laude honors in Psychology/Neuroscience

2013

Dean's List

2009 – 2013

PEER-REVIEWED PUBLICATIONS

Schreiner, D.C., Brudner, S.N., Li, A., Pearson, J.M., Mooney, R.D. (2026). A synaptic locus of song learning. *Nature*, 655(8124), 997–1005. <https://doi.org/10.1038/s41586-026-10510-x>

Qi, J., **Schreiner, D.C.**, Martinez, M., Pearson, J.M., Mooney, R.D. (2025). Dual neuromodulatory dynamics underlie birdsong learning. *Nature*, 641(8063), 690–698. <https://doi.org/10.1038/s41586-025-08694-9>

Schreiner, D.C., Wright, A., Baltz, E.T., Wang, T., Cazares, C., Gremel, C.M. (2023). Chronic alcohol exposure alters action control via hyperactive premotor corticostriatal activity. *Cell Reports*, 42(7), 112675. <https://doi.org/10.1016/j.celrep.2023.112675>

Cazares, C., **Schreiner, D.C.**, Valencia, M.L., Gremel, C.M. (2022). Orbitofrontal cortex populations are differentially recruited to support actions. *Current Biology*, 32(21), 4676–4687.e5. <https://doi.org/10.1016/j.cub.2022.09.022>

Schreiner, D.C., Cazares, C., Renteria, R., Gremel, C.M. (2022). Information normally considered task-irrelevant drives decision-making and affects premotor circuit recruitment. *Nature Communications*, 13(1), 2134. <https://doi.org/10.1038/s41467-022-29807-2>

Schreiner, D.C., Yalcinbas, E.A., Gremel, C.M. (2021). A push for examining subjective experience in value-based decision-making. *Current Opinion in Behavioral Sciences*, 41, 45–49. <https://doi.org/10.1016/j.cobeha.2021.03.020>

Cazares, C., **Schreiner, D.C.**, Gremel, C.M. (2021). Different effects of alcohol exposure on action and outcome related orbitofrontal cortex activity. *eNeuro*, 8(2), ENEURO.0052-21.2021. <https://doi.org/10.1523/ENEURO.0052-21.2021>

Renteria, R., Cazares, C., Baltz, E.T., **Schreiner, D.C.**, Yalcinbas, E.A., Steinkellner, T., Hnasko, T.S., Gremel, C.M. (2021). Mechanisms for differential recruitment of orbitostriatal transmission during actions and outcomes following chronic alcohol exposure. *eLife*, 10, e67065. <https://doi.org/10.7554/eLife.67065>

Schreiner, D.C., Renteria, R., Gremel, C.M. (2019). Fractionating the all-or-nothing definition of goal-directed and habitual decision-making. *Journal of Neuroscience Research*, 98(6), 998–1006. <https://doi.org/10.1002/jnr.24545>

Schreiner, D.C., Gremel, C.M. (2018). Orbital frontal cortex projections to secondary motor cortex mediate exploitation of learned rules. *Scientific Reports*, 8, 10979. <https://doi.org/10.1038/s41598-018-29285-x>

*Kavanagh, K.A., ***Schreiner, D.C.**, Levis, S.C., O'Neill, C.E., Bachtell, R.K. (2015). Role of adenosine receptor subtypes in methamphetamine reward and reinforcement. *Neuropharmacology*, 89, 265–273. <https://doi.org/10.1016/j.neuropharm.2014.09.030> ***Equal contribution**

O'Neill, C.E., Levis, S.C., **Schreiner, D.C.**, Amat, J., Maier, S.F., Bachtell, R.K. (2015). Effects of adolescent caffeine consumption on cocaine sensitivity. *Neuropsychopharmacology*, 40(4), 813–821. <https://doi.org/10.1038/npp.2014.278>

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INVITED SEMINARS AND ORAL PRESENTATIONS

Emerging Scientist Seminar, Max Planck Institute for Biological Intelligence	June 2027
Dendrites: Molecules, Structure and Function Gordon Conference	March 2027
NEURO-ROOT Seminar, University of Iowa, Neuroscience and Pharmacology	Oct. 2026
Duke Neurobiology Annual Retreat	Sept. 2026
University of Colorado Anschutz School of Medicine, Pharmacology	July 2026
Neural Mechanisms of Acoustic Communication Gordon Conference	June 2026
Invited Postdoc Seminar, Mount Sinai, Neuroscience	April 2026
Invited GPCR Postdoc Seminar, Vanderbilt University, Pharmacology	March 2026
Future of Neuroethology Webinar, International Society for Neuroethology	March 2026
Penn State University, Biology	March 2026
University of Wisconsin–Madison, Integrative Biology	Feb. 2026
Basal Ganglia Gordon Conference	Feb. 2026
Basal Ganglia Gordon Seminar	Feb. 2026
University of Oxford, Biology	Feb. 2026
Brain Talks on the Isthmus, University of Wisconsin–Madison, Psychology	Dec. 2025
Max Planck Institute for Biological Cybernetics Symposium	Nov. 2025
Lake Conference on Neural Coding and Dynamics	Oct. 2025
University of Southern Denmark, Biology	June 2025
Virtual Songbird Satellite Symposium	April 2025
Swedish Basal Ganglia Society Annual Meeting	Dec. 2024
NIH Outstanding Scholar in Neuroscience Symposium	Nov. 2023
Duke Neurobiology Annual Retreat	Sept. 2023
Research Society on Alcoholism Enoch Gordis Award Presentation	June 2021

POSTER ABSTRACTS

Schreiner, D.C., Li, A., Mercer, A.A., Pearson, J.M., Mooney, R.D. (2026). From synaptic locus towards molecular mechanism of song learning. Soc. for Neuro. Abst. 6057.

Schreiner, D.C., Brudner, S.N., Li, A., Pearson, J.M., Mooney, R.D. (2026). A synaptic locus of song learning. Gordon Neural Mech. of Acoustic Communication.

Schreiner, D.C., Brudner, S.N., Li, A., Pearson, J.M., Mooney, R.D. (2026). A corticostriatal locus of birdsong learning. Gordon Basal Ganglia.

Schreiner, D.C., Brudner, S.N., Li, A., Pearson, J.M., Mooney, R.D. (2025). A synaptic locus of song learning. Soc. for Neuro. Abst. 174.23.

Schreiner, D.C., Brudner, S.N., Li, A., Pearson, J.M., Mooney, R.D. (2025). A synaptic locus of song learning. Gordon Neuroethology.

Schreiner, D.C., Brudner, S.N., Pearson, J.M., Mooney, R.D. (2025). A synaptic locus of song learning. Janelia Grounding Cognition in Mechanistic Insight conference.

O’Gorman, E., **Schreiner, D.C.**, Mooney, R.D., Pearson, J.M. (2025). Hacking vocal learning with deep learning: flexible real-time perturbation of zebra finch song. Computational and Systems Neuroscience.

Schreiner, D.C., Brudner, S.N., Pearson, J.M., Mooney, R.D. (2024). A synaptic locus of song learning. Annual BRAIN Initiative Conference. Abst. 613.

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Schreiner, D.C., Brudner, S.N., Pearson, J.M., Mooney, R.D. (2024). A synaptic locus of song learning. *Gordon Neural Mech. of Acoustic Communication*. Abst. 51.

O’Gorman, E., **Schreiner, D.C.**, Mooney, R.D., Pearson, J.M. (2024). Hacking vocal learning with deep learning: flexible real-time perturbation of zebra finch song. *Gordon Neural Mech. of Acoustic Communication*.

Schreiner, D.C., Brudner, S.N., Pearson, J.M., Mooney, R.D. (2024). A synaptic locus of song learning. *Gordon Basal Ganglia*. Abst. 32.

Schreiner, D.C., Brudner, S.N., Pearson, J.M., Mooney, R.D. (2024). A synaptic locus of song learning. *Computational and Systems Neuroscience*. Abst. 2-070.

Schreiner, D.C., Brudner, S.N., Pearson, J.M., Mooney, R.D. (2023). Basal ganglia-dependent expression of recent song learning in the juvenile finch. *Computational and Systems Neuroscience*. Abst. 2-003.

Schreiner, D.C., Brudner, S.N., Pearson, J.M., Mooney, R.D. (2022). Expression of recent learning requires basal ganglia activity during juvenile song copying. *Soc. for Neuro*. Abst. 135.24.

Schreiner, D.C., Renteria, R., Gremel, C.M. (2021). Chronic ethanol exposure increases premotor cortex activity and its control over behavioral flexibility. *Res. Soc. on Alcoholism Annual Meeting*. Abst. 553.

Schreiner, D.C., Gremel, C.M. (2021). Subjective experience carried by premotor output to striatum supports decision-making. *Computational and Systems Neuroscience*. Abst. 2-032.

Schreiner, D.C., Gremel, C.M. (2020). Premotor output to striatum carries action history information to support ongoing decision-making. *Computational and Systems Neuroscience*. Abst. 1-086.

Schreiner, D.C., Renteria, R., Gremel, C.M. (2019). Aberrantly persistent reward seeking following alcohol dependence and its premotor cortical determinants. *Res. Soc. on Alcoholism Annual Meeting*. Abst. 269.

Schreiner, D.C., Gremel, C.M. (2018). The role of the secondary motor cortex in feedback integration to guide decision-making. *Soc. for Neuro*. Abst. 507.09.

Rossa, M., Lovelett, J., **Schreiner, D.C.**, Spencer, B., Kirchgessner, M. (2018). NeuwriteSD: Building skills in science communication through writing. *Soc. for Neuro*. Abst. 026.19SU.

Cazares, C., **Schreiner, D.C.**, Gremel, C.M. (2018). Ethanol modulation of orbitofrontal cortex activity during goal-directed action. *Res. Soc. on Alcoholism Annual Meeting*.

Renteria, R., Baltz, E.T., **Schreiner, D.C.**, Gremel, C.M. (2018). Chronic ethanol exposure alters dorsomedial striatal D1 receptor function disrupting goal-directed actions. *Res. Soc. on Alcoholism Annual Meeting*.

Schreiner, D.C., Perez, J., Gremel, C.M. (2018). The contribution of rodent secondary motor cortex to feedback guided actions. *Annual Winter Conference on Brain Research*.

Perez, J., **Schreiner, D.C.**, Gremel, C.M. (2017). Behavioral feedback in mice. *Society for Advancement of Chicanos and Native Americans in Science*, Salt Lake City, Utah.

Schreiner, D.C., Renteria, R., Baltz, E.T., Gremel, C.M. (2017). The effect of prior chronic intermittent ethanol and withdrawal on action generalization. *Res. Soc. on Alcoholism Annual Meeting*. Abst. 054.

Schreiner, D.C., Gremel, C.M. (2016). Role of mouse premotor cortex in action generalization and goal-directed behavior. *Soc. for Neuro*. Abst. 7805.

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Haynes, N., O'Neill, C.E., Levis, S.C., **Schreiner, D.**, Stafford, J., Bachtell, R.K. (2015). Subpopulations of adenosine A2A receptors in the nucleus accumbens play different roles in cocaine seeking. Soc. for Neuro. Abst. 51.15.

O'Neill, C.E., Levis, S.C., **Schreiner, D.**, Bachtell, R.K. (2014). Blockade of presynaptic and postsynaptic adenosine A2A receptors produces bi-directional effects on cocaine seeking. ACNP Annual Meeting.

O'Neill, C.E., Levis, S.C., **Schreiner, D.**, Bachtell, R.K. (2014). Opposing effects of pre and postsynaptic adenosine A2A receptor blockade on cocaine seeking. Soc. for Neuro. Abst. 430.09.

Schreiner, D.C., Bachtell, R.K. (2013). Assessing cocaine addiction vulnerability and neurobiological correlates within the nigrostriatal pathway using a novel model of Dopamine D2 autoreceptor sensitivity. Howard Hughes Medical Institute Biomedical Sciences Grant Poster Session.

TEACHING

Graduate Teaching, Duke University

Co-designed and co-instructed *Navigating the Postdoctoral Pathway*

Summer 2024

Guest Lecturer, *Concepts in Neurobiology II*, Learning and Memory: Birdsong

Spring 2024, 2025

Undergraduate Teaching, University of California San Diego

- 8 teaching assistant appointments
- Delivered at least one guest lecture in every appointment; 95% of 89 student evaluations recommended me as a TA; all faculty supervisors rated my performance "Excellent"

Clinical Psychology

Spring 2020

Cognitive Neuroscience

Spring 2019

Criminology

Fall 2016, Fall 2017, Fall 2018

Behavior Modification

Winter 2018

Drugs and Behavior

Winter 2017

Impulse Control Disorders

Winter 2016

MENTORSHIP

Research Mentor, Duke University – 5 trainees

Graduate students

Jiaxuan Qi (first author, *Nature*, 2025; now LSRF Postdoctoral Fellow, Witten Lab, Princeton University)

Shuoyi Li (current Ph.D. student)

Kate Kaplan (current Ph.D. student)

Undergraduates

Amanda Li (co-author, *Nature*, 2026)

Autumn Wetzell (subsequently Research Technician, Mooney Lab)

Research Mentor, UC San Diego – 4 undergraduate trainees

Joselyn Perez (subsequently post-baccalaureate researcher, IUPUI; Ph.D., University of Connecticut, 2026)

Ni "Jenny" Zhen (subsequently Lab Manager, University of Chicago; now Stripe)

Dolly Mizner (subsequently trained in applied behavior analysis; now Board Certified Behavior Analyst)

Esra Elhendy

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SCIENCE COMMUNICATION, OUTREACH, AND SERVICE

Mentor, Brain Science Career Day for Historically Black Colleges and Universities 2024 – 2026
Reviewed CVs and conducted mock interviews for HBCU students pursuing graduate or medical school

Co-Editor-in-Chief, NeuWrite San Diego (NeuWriteSD.org) 2019 – 2021
Led publication of 91 public-facing neuroscience articles, including 3 authored personally; 168,000 views, with 27% originating in Spanish-speaking countries

Steering Committee, NeuWrite San Diego 2015 – 2021
Before becoming Co-Editor-in-Chief, helped lead publication of 187 articles, including 8 authored personally; helped establish a Spanish-language translation initiative; 250,000 views with 15% originating from Spanish-speaking countries; secured three Society for Neuroscience outreach grants (\$4,500 total)

Mentor, Graduate Application and Mentorship Program 2020 – 2021
Mentored prospective graduate students through the application process

Organizing Committee, Communicating Science Conference San Diego 2015 – 2019
Regional conference that trained graduate students and postdocs in science communication

Organizing Committee, National Communicating Science Conference 2018 – 2019
National conference that trained graduate students and postdocs in science communication

Organizing Committee, Taste of Science/Pint of Science 2016 – 2019
Helped organize public science talks bringing researchers into community venues

REVIEWING

Ad hoc reviewer: *Cell Reports*, *Current Biology*, *Science Advances*, *Current Opinion in Neurobiology*, *Nature Communications*, *Cosyne*, Graduate Women in Science National Fellowship Program

PROFESSIONAL DEVELOPMENT

Neuromatch Computational Neuroscience Summer Course Summer 2022
Introduction to College Teaching Fall 2019

PROFESSIONAL ORGANIZATIONS

Science Communicators of North Carolina 2022 – Present
International Society for Neuroethology 2021 – Present
Faculty for Undergraduate Neuroscience 2019 – Present
Society for Neuroscience 2015 – Present
San Diego Science Writers Association 2019 – 2021
Research Society on Alcoholism 2016 – 2021

OTHER ACHIEVEMENTS

Thru-hiked the 2,185-mile Appalachian Trail from Georgia to Maine 2014

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REFERENCES

On Request